

The Complete File-Search Program

Let's begin by taking a look at the complete source code for the recursive file-search program. The rest of this chapter explains each section of code individually. Copy the source code for the file-search program to a file named *fileFinder.py*:

```
import os

def hasEvenByteSize(fullFilePath):
    """Returns True if fullFilePath has an even size in bytes,
    otherwise returns False."""
    fileSize = os.path.getsize(fullFilePath)
    return fileSize % 2 == 0

def hasEveryVowel(fullFilePath):
    """Returns True if the fullFilePath has a, e, i, o, and u,
    otherwise returns False."""
    name = os.path.basename(fullFilePath).lower()
    return ('a' in name) and ('e' in name) and ('i' in name) and ('o' in name) and ('u' in name)

def walk(folder, matchFunc):
    """Calls the match function with every file in the folder and its
    subfolders. Returns a list of files that the match function
    returned True for."""
    matchedFiles = [] # This list holds all the matches.
    folder = os.path.abspath(folder) # Use the folder's absolute path.

    # Loop over every file and subfolder in the folder:
    for name in os.listdir(folder):
        filepath = os.path.join(folder, name)
        if os.path.isfile(filepath):
            # Call the match function for each file:
            if matchFunc(filepath):
                matchedFiles.append(filepath)
        elif os.path.isdir(filepath):
            # Recursively call walk for each subfolder, extending
            # the matchedFiles with their matches:
```